

Chapter 20

General-Purpose Registers (GPR_1)

20.1 Chip-specific GPR_1 information

20.1.1 GPR_1 instances

This chip supports one instance of the GPR_1 module.

Register implementation for S32K511.

New Registers in S32K511

The following registers are applicable to S32K511 device.

Table 70. New Registers in S32K511

Register Name	Offset Address	Access	Reset Value	Description
GPR_RWF_10	2028h	RW	0000_0000h	New register in S32K511
GPR_RWF_11	202Ch	RW	0000_0000h	New register in S32K511
GPR_RWF_12	2030h	RW	0000_0000h	New register in S32K511
GPR_RWF_9	2024h	RW	0000_0000h	New register in S32K511

Register details:

20.1.2 Read Write FUNC 7 (GPR_RWF_9)

Offset

Register	Offset
GPR_RWF_9	2024h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LOCK_	LOCK_	Reserved													
W	EN1	EN0														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LPSP	LPSP	LPSP2_EN_PCS_DELAY								LPSP2_PCS_EN_DELAY					
W	_2...	_2...														
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 LOCK_EN1	Lock enable. If LOCK_EN1 and LOCK_EN0 both are 0 then register can be written. If LOCK_EN1 is 0 and LOCK_EN0 is 1 then register can be written. If LOCK_EN1 is 1 and LOCK_EN0 is 0 then register is temporarily protected. If LOCK_EN1 and LOCK_EN0 both are 1 then this register is locked (read-only) until the next reset.
30 LOCK_EN0	Lock enable. For detailed description of LOCK_EN0 and LOCK_EN1 bit please see LOCK_EN1 section.
29-16 —	Reserved
15 LPSP1_2_EN_P OL	Decoder Enable Polarity This is select input of mux which select the polarity of Decoder enable signal 0b - Non-Inverted Decoder enable signal 1b - Inverted Decoder enable signal
14 LPSP1_2_PCS_ POL	Programmable Chip Select Polarity This is select input of mux which select the polarity of PCS signal 0b - Non-Inverted PCS value 1b - Inverted PCS value
13-7 LPSP1_2_EN_P CS_DELAY	Programmable Chip Select - Enable delay These bits are using as inserting delay in between PCS signal and decoder enable signal to prevent glitch propogation in system through PCS signal. It's time between PCS and decoder enable after disabling the decoder enable and before transition the PCS.
6-0 LPSP1_2_PCS_ EN_DELAY	Programmable Chip Select - Enable delay These bits are using as inserting delay in between PCS signal and decoder enable signal to prevent glitch propogation in system through PCS signal. It provide time to settle PCS signal and propogate glichless PCS signal to the system.

20.1.3 Read Write FUNC 7 (GPR_RWF_10)**Offset**

Register	Offset
GPR_RWF_10	2028h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
LPSPi_3_PCS_EN_DELAY	These bits are using as inserting delay in between PCS signal and decoder enable signal to prevent glitch propogation in system through PCS signal. It provide time to settle PCS signal and propogate glichless PCS signal to the system.

20.1.4 Read Write FUNC 7 (GPR_RWF_11)

Offset

Register	Offset
GPR_RWF_11	202Ch

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	LOCK_EN1	LOCK_EN0	Reserved													
W	LOCK_EN1	LOCK_EN0	Reserved													
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	LPSPi_4...	LPSPi_4...	LPSPi_4_EN_PCS_DELAY								LPSPi_4_PCS_EN_DELAY					
W	LPSPi_4...	LPSPi_4...	LPSPi_4_EN_PCS_DELAY								LPSPi_4_PCS_EN_DELAY					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 LOCK_EN1	Lock enable. If LOCK_EN1 and LOCK_EN0 both are 0 then register can be written. If LOCK_EN1 is 0 and LOCK_EN0 is 1 then register can be written. If LOCK_EN1 is 1 and LOCK_EN0 is 0 then register is temporarily protected. If LOCK_EN1 and LOCK_EN0 both are 1 then this register is locked (read-only) until the next reset.
30 LOCK_EN0	Lock enable. For detailed description of LOCK_EN0 and LOCK_EN1 bit please see LOCK_EN1 section.
29-16	Reserved

Table continues on the next page...

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Field	Function
—	
15 LPSP1_4_EN_P OL	Decoder Enable Polarity This is select input of mux which select the polarity of Decoder enable signal 0b - Non-Inverted Decoder enable signal 1b - Inverted Decoder enable signal
14 LPSP1_4_PCS_ POL	Programmable Chip Select Polarity This is select input of mux which select the polarity of PCS signal 0b - Non-Inverted PCS value 1b - Inverted PCS value
13-7 LPSP1_4_EN_P CS_DELAY	Programmable Chip Select - Enable delay These bits are using as inserting delay in between PCS signal and decoder enable signal to prevent glitch propagation in system through PCS signal. It's time between PCS and decoder enable after disabling the decoder enable and before transition the PCS.
6-0 LPSP1_4_PCS_ EN_DELAY	Programmable Chip Select - Enable delay These bits are using as inserting delay in between PCS signal and decoder enable signal to prevent glitch propagation in system through PCS signal. It provide time to settle PCS signal and propagate glitchless PCS signal to the system.

20.1.5 Read Write FUNC 7 (GPR_RWF_12)

Offset

Register	Offset
GPR_RWF_12	2030h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
LPSP1_5_PCS_EN_DELAY	These bits are using as inserting delay in between PCS signal and decoder enable signal to prevent glitch propagation in system through PCS signal. It provide time to settle PCS signal and propagate glitchless PCS signal to the system.

20.1.6

Registers with Attribute Changes in S32K511

The following registers exist in both S32K566 and S32K511 but have changed offset, access type, or reset values.

Table 71. Registers with Modified Attributes in S32K511

Register	S32K566 Offset	S32K566 Access	S32K566 Reset	S32K511 Offset	S32K511 Access	S32K511 Reset	Change Summary
GPR_ROF_0	2800h	RW	0000_000Fh	2800h	RW	0000_0000h	Reset value changed from 0000_000Fh to 0000_0000h
GPR_RWF_0	2000h	RW	0000_0200h	2000h	RW	0000_0000h	Reset value changed from 0000_0200h to 0000_0000h
GPR_RWF_3	200Ch	RW	0000_0000h	200Ch	R	0000_0000h	Access changed from RW to R
GPR_RWF_5	2014h	RW	2000_0000h	2014h	RW	0000_0000h	Reset value changed from 2000_0000h to 0000_0000h

The following bit fields are applicable in S32K511 registers.

Table 72. New Bit Fields in S32K511

Register	Bit Field	Field Name	Description for S32K511
GPR_ROF_0	3-0	Reserved	Reserved
GPR_ROF_1	9-6	Reserved	Reserved
GPR_ROF_1	4	INIT_OCM_DONE	Reserved
GPR_RWD_10	25-24	Reserved	Reserved
GPR_RWD_10	16-13	Reserved	Reserved
GPR_RWD_10	6-0	Reserved	Reserved
GPR_RWD_10	8	Reserved	Reserved
GPR_RWD_11	4-3	Reserved	Reserved
GPR_RWD_13	25-24	Reserved	Reserved

Table continues on the next page...

Table 72. New Bit Fields in S32K511 (continued)

Register	Bit Field	Field Name	Description for S32K511
GPR_RWD_7	6-0	Reserved	Reserved
GPR_RWD_8	4-3	Reserved	Reserved
GPR_RWF_1	1-0	Reserved	Reserved
GPR_RWF_1	2	Reserved	Reserved
GPR_RWF_1	3	Reserved	Reserved
GPR_RWF_1	5	Reserved	Reserved
GPR_RWF_1	8-7	Reserved	Reserved
GPR_RWF_1	11-10	Reserved	Reserved
GPR_RWF_1	14-13	Reserved	Reserved
GPR_RWF_3	31-0	Reserved	Reserved
GPR_RWF_5	29-17	Reserved	Reserved
GPR_RWF_9	29-16	Reserved	Reserved

20.2 Overview

GPR1 is a group of user programmable registers, which configures and captures status for multiple functions on this chip. Functional descriptions of register bits are provided in subsequent sections of this chapter.

20.3 Memory Map and register definition

This section includes the GPR1 module memory map and detailed descriptions of all registers.

20.3.1 GPR1 register descriptions

20.3.1.1 GPR1 memory map

GPR_1 base address: 4045_8000h

Offset	Register	Width (In bits)	Access	Reset value
1000h	Read Write DEST 0 (GPR_RWD_0)	32	RW	0000_0000h
1004h	Read Write DEST 1 (GPR_RWD_1)	32	RW	0000_0000h
1008h	Read Write DEST 2 (GPR_RWD_2)	32	RW	0000_0000h
100Ch	Read Write DEST 3 (GPR_RWD_3)	32	RW	0000_0004h
1010h	Read Write DEST 4 (GPR_RWD_4)	32	RW	0000_0000h
1014h	Read Write DEST 5 (GPR_RWD_5)	32	RW	0000_0000h
1018h	Read Write DEST 6 (GPR_RWD_6)	32	RW	0000_0000h
101Ch	Read Write DEST 7 (GPR_RWD_7)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
10B8h	Read Write DEST 42 (GPR_RWD_46)	32	RW	0000_0000h
1800h	Read Only DEST 0 (GPR_ROD_0)	32	R	0000_0000h
1804h	Read Only DEST 1 (GPR_ROD_1)	32	R	0000_0000h
1808h	Read Only DEST 2 (GPR_ROD_2)	32	R	0000_0000h
180Ch	Read Only DEST 3 (GPR_ROD_3)	32	R	0000_0000h
1810h	Read Only DEST 4 (GPR_ROD_4)	32	RW	0000_0000h
1814h	Read Only DEST 5 (GPR_ROD_5)	32	RW	0000_0000h
1818h	Read Only DEST 6 (GPR_ROD_6)	32	RW	0000_0000h
181Ch	Read Only DEST 7 (GPR_ROD_7)	32	RW	0000_0000h
1820h	Read Only DEST 8 (GPR_ROD_8)	32	RW	0000_0000h
1824h	Read Only DEST 9 (GPR_ROD_9)	32	RW	0000_0000h
1848h	Read Only DEST 4 (GPR_ROD_11)	32	R	0000_0000h
184Ch	Read Only DEST 5 (GPR_ROD_12)	32	RW	0000_0000h
2000h	Read Write FUNC 0 (GPR_RWF_0)	32	RW	0000_0200h
2004h	Read Write FUNC 1 (GPR_RWF_1)	32	RW	0000_0000h
2008h	Read Write FUNC 2 (GPR_RWF_2)	32	RW	0002_0000h
200Ch	Read Write FUNC 3 (GPR_RWF_3)	32	RW	0000_0000h
2010h	Read Write FUNC 4 (GPR_RWF_4)	32	RW	0000_0001h
2014h	Read Write FUNC 5 (GPR_RWF_5)	32	RW	2000_0000h
2018h	Read Write FUNC 6 (GPR_RWF_6)	32	RW	0000_0000h
201Ch	Read Write FUNC 7 (GPR_RWF_7)	32	RW	0000_0000h
2020h	Read Write FUNC 3 (GPR_RWF_8)	32	RW	0000_0000h
2800h	Read Only FUNC 0 (GPR_ROF_0)	32	RW	0000_000Fh
2804h	Read Only FUNC 1 (GPR_ROF_1)	32	R	0000_0000h
2808h	Read Only FUNC 2 (GPR_ROF_2)	32	R	0000_0001h

20.3.1.2 Read Write DEST 0 (GPR_RWD_0)

Offset

Register	Offset
GPR_RWD_0	1000h

Table continued from the previous page...

Field	Function
MUX_SEL15	This bit overrides the functionality selected for GPIO81 to always select EOUT functionality. 0b - Functions on GPIO81 can be selected as per S32K56x_IOMUX.xlsx. 1b - EOUT will be selected.
14 MUX_SEL14	EOUT select. Mux select for selection between EOUT0 and EOUT1 for GPIO81 0b - EOUT0 will be selected 1b - EOUT1 will be selected
13 MUX_SEL13	EOUT Override. This bit overrides the functionality selected for GPIO80 to always select EOUT functionality. 0b - Functions on GPIO80 can be selected as per S32K56x_IOMUX.xlsx. 1b - EOUT will be selected.
12 MUX_SEL12	EOUT select. Mux select for selection between EOUT0 and EOUT1 for GPIO80 0b - EOUT0 will be selected 1b - EOUT1 will be selected
11-6 OTA_NUM_FIXED_LPART	OTA feature support for MRAM. To support OTA feature in MRAM. Refer OTA section in MRAMC chapter for more details. 00_0000b - Full swap. 00_0100b - Partial swap. 01_0000b - All fixed - no OTA.
5 OTA_SEL	OTA Select for MRAM large partitions. Controls swap of non fixed large partitions when either in full or partial OTA modes. 0b - No swap 1b - Swap non-fixed large partitions
4-0 —	Reserved

20.3.1.3 Read Write DEST 1 (GPR_RWD_1)

Offset

Register	Offset
GPR_RWD_1	1004h

Table continued from the previous page...

Field	Function
CM7_3_FAULT	Initiate software fault to FCCU. 0b - Fault to FCCU will not be generated. 1b - Fault to FCCU will be generated.
2 CM7_2_FAULT	Software fault to CM7_2 Initiate software fault to FCCU. 0b - Fault to FCCU will not be generated. 1b - Fault to FCCU will be generated.
1 CM7_1_FAULT	Software fault to CM7_1 Initiate software fault to FCCU. 0b - Fault to FCCU will not be generated. 1b - Fault to FCCU will be generated.
0 CM7_0_FAULT	Software fault to CM7_0 Initiate software fault to FCCU. 0b - Fault to FCCU will not be generated. 1b - Fault to FCCU will be generated.

20.3.1.4 Read Write DEST 2 (GPR_RWD_2)

Offset

Register	Offset
GPR_RWD_2	1008h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

20.3.1.5 Read Write DEST 3 (GPR_RWD_3)

Offset

Register	Offset
GPR_RWD_3	100Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved	CLOCKOUT_RUN_MUX_SEL							CLOCKOUT_RUN_DIV_VAL							CLOCKOUT_RUN_CLK_EN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Fields

Field	Function
31-15 —	Reserved
14-9 CLOCKOUT_RUN_MUX_SEL	Mux sel value of CLOCKOUT_RUN. Mux sel value of CLOCKOUT_RUN.
8-1 CLOCKOUT_RUN_DIV_VAL	Divider value of CLOCKOUT_RUN. Divider value of CLOCKOUT_RUN.
0 CLOCKOUT_RUN_CLK_EN	Enable/Disable CLOCKOUT_RUN to be exposed. Enable/Disable CLOCKOUT_RUN to be exposed. 0b - Disabled. 1b - Enabled.

20.3.1.6 Read Write DEST 4 (GPR_RWD_4)

Offset

Register	Offset
GPR_RWD_4	1010h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserv	Reserv	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	ed	ed	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 —	Reserved
30 —	Reserved
29 DEBUG_DISABLE LE29	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - LCU1 enters debug mode when CM7_0 enters debug mode. 1b - LCU1 remains functional and is not impacted when CM7_0 enters debug mode.
28 DEBUG_DISABLE LE28	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - LCU0 enters debug mode when CM7_0 enters debug mode. 1b - LCU0 remains functional and is not impacted when CM7_0 enters debug mode.
27	DEBUG_DISABLE

Table continues on the next page...

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Field	Function
DEBUG_DISAB LE4	0b - CPE_VFCCU enters debug mode when CM7_0 enters debug mode. 1b - CPE_VFCCU remains functional and is not impacted when CM7_0 enters debug mode.
3 DEBUG_DISAB LE3	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - CPE_SWT1 enters debug mode when CM7_0 enters debug mode. 1b - CPE_SWT1 remains functional and is not impacted when CM7_0 enters debug mode.
2 DEBUG_DISAB LE2	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - CPE_SWT0 enters debug mode when CM7_0 enters debug mode. 1b - CPE_SWT0 remains functional and is not impacted when CM7_0 enters debug mode.
1 DEBUG_DISAB LE1	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when CM7_0 enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when CM7_0 enters debug mode.
0 DEBUG_DISAB LE0	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when CM7_0 enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when CM7_0 enters debug mode.

20.3.1.7 Read Write DEST 5 (GPR_RWD_5)

Offset

Register	Offset
GPR_RWD_5	1014h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - DSPI0 remains functional and is not impacted when CM7_0 enters debug mode.
2 DEBUG_DISAB LE2	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - VFCCU3 enters debug mode when CM7_0 enters debug mode. 1b - VFCCU3 remains functional and is not impacted when CM7_0 enters debug mode.
1 DEBUG_DISAB LE1	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - VFCCU2 enters debug mode when CM7_0 enters debug mode. 1b - VFCCU2 remains functional and is not impacted when CM7_0 enters debug mode.
0 DEBUG_DISAB LE0	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when CM7_0 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when CM7_0 enters debug mode.

20.3.1.8 Read Write DEST 6 (GPR_RWD_6)

Offset

Register	Offset
GPR_RWD_6	1018h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															DEBU
W																G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
1 DEBUG_DISABLE1	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - STM0 enters debug mode when CM7_0 enters debug mode. 1b - STM0 remains functional and is not impacted when CM7_0 enters debug mode.
0 DEBUG_DISABLE0	DEBUG_DISABLE Debug disable bit for CM7_0. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_3 enters debug mode when CM7_0 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when CM7_0 enters debug mode.

20.3.1.9 Read Write DEST 7 (GPR_RWD_7)

Offset

Register	Offset
GPR_RWD_7	101Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved	Reserved	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...
W	Reserved	Reserved	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...
W	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 —	Reserved

Table continues on the next page...

20.3.1.10 Read Write DEST 8 (GPR_RWD_8)

Offset

Register	Offset
GPR_RWD_8	1020h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 DEBUG_DISAB LE31	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_2 enters debug mode when CM7_1 enters debug mode. 1b - PIT_RTI_2 remains functional and is not impacted when CM7_1 enters debug mode.
30 DEBUG_DISAB LE30	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_1 enters debug mode when CM7_1 enters debug mode. 1b - PIT_RTI_1 remains functional and is not impacted when CM7_1 enters debug mode.
29 DEBUG_DISAB LE29	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_0 enters debug mode when CM7_1 enters debug mode. 1b - PIT_RTI_0 remains functional and is not impacted when CM7_1 enters debug mode.
28	DEBUG_DISABLE

Table continues on the next page...

Table continued from the previous page...

Field	Function
DEBUG_DISAB LE5	0b - eMIOS0 enters debug mode when CM7_1 enters debug mode. 1b - eMIOS0 remains functional and is not impacted when CM7_1 enters debug mode.
4 DEBUG_DISAB LE4	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - DSPI1 enters debug mode when CM7_1 enters debug mode. 1b - DSPI1 remains functional and is not impacted when CM7_1 enters debug mode.
3 DEBUG_DISAB LE3	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - DSPI0 enters debug mode when CM7_1 enters debug mode. 1b - DSPI0 remains functional and is not impacted when CM7_1 enters debug mode.
2 DEBUG_DISAB LE2	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - VFCCU3 enters debug mode when CM7_1 enters debug mode. 1b - VFCCU3 remains functional and is not impacted when CM7_1 enters debug mode.
1 DEBUG_DISAB LE1	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - VFCCU2 enters debug mode when CM7_1 enters debug mode. 1b - VFCCU2 remains functional and is not impacted when CM7_1 enters debug mode.
0 DEBUG_DISAB LE0	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when CM7_1 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when CM7_1 enters debug mode.

20.3.1.11 Read Write DEST 9 (GPR_RWD_9)

Offset

Register	Offset
GPR_RWD_9	1024h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - STM3 remains functional and is not impacted when CM7_1 enters debug mode.
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - STM2 enters debug mode when CM7_1 enters debug mode. 1b - STM2 remains functional and is not impacted when CM7_1 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - STM1 enters debug mode when CM7_1 enters debug mode. 1b - STM1 remains functional and is not impacted when CM7_1 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - STM0 enters debug mode when CM7_1 enters debug mode. 1b - STM0 remains functional and is not impacted when CM7_1 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM7_1. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_3 enters debug mode when CM7_1 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when CM7_1 enters debug mode.

20.3.1.12 Read Write DEST 10 (GPR_RWD_10)

Offset

Register	Offset
GPR_RWD_10	1028h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - CPE_SWT0 remains functional and is not impacted when CM7_2 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM7_2. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when CM7_2 enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when CM7_2 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM7_2. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when CM7_2 enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when CM7_2 enters debug mode.

20.3.1.13 Read Write DEST 11 (GPR_RWD_11)

Offset

Register	Offset
GPR_RWD_11	102Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM7_2. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when CM7_2 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when CM7_2 enters debug mode.

20.3.1.14 Read Write DEST 12 (GPR_RWD_12)

Offset

Register	Offset
GPR_RWD_12	1030h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															DEBU
W																G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-17 —	Reserved
16 DEBUG_DISABLE LE16	DEBUG_DISABLE Debug disable bit for CM7_2. Set this bit 1 to disable the debug of module. 0b - SENT3 enters debug mode when CM7_2 enters debug mode. 1b - SENT3 remains functional and is not impacted when CM7_2 enters debug mode.

Table continues on the next page...

Table continued from the previous page...

Field	Function
DEBUG_DISAB LE0	0b - PIT_RTI_3 enters debug mode when CM7_2 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when CM7_2 enters debug mode.

20.3.1.15 Read Write DEST 13 (GPR_RWD_13)

Offset

Register	Offset
GPR_RWD_13	1034h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserv ed	Reserv ed	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...	DEBU G_D...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 —	Reserved
30 —	Reserved
29 DEBUG_DISAB LE29	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - LCU1 enters debug mode when CM7_3 enters debug mode.

Table continues on the next page...

Table continued from the previous page...

Field	Function
5 DEBUG_DISABLE LE5	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - CPE_PIT0 enters debug mode when CM7_3 enters debug mode. 1b - CPE_PIT0 remains functional and is not impacted when CM7_3 enters debug mode.
4 DEBUG_DISABLE LE4	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - CPE_VFCCU enters debug mode when CM7_3 enters debug mode. 1b - CPE_VFCCU remains functional and is not impacted when CM7_3 enters debug mode.
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - CPE_SWT1 enters debug mode when CM7_3 enters debug mode. 1b - CPE_SWT1 remains functional and is not impacted when CM7_3 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - CPE_SWT0 enters debug mode when CM7_3 enters debug mode. 1b - CPE_SWT0 remains functional and is not impacted when CM7_3 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when CM7_3 enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when CM7_3 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when CM7_3 enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when CM7_3 enters debug mode.

20.3.1.16 Read Write DEST 14 (GPR_RWD_14)

Offset

Register	Offset
GPR_RWD_14	1038h

Table continued from the previous page...

Field	Function
	1b - DSPI1 remains functional and is not impacted when CM7_3 enters debug mode.
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - DSPI0 enters debug mode when CM7_3 enters debug mode. 1b - DSPI0 remains functional and is not impacted when CM7_3 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - VFCCU3 enters debug mode when CM7_3 enters debug mode. 1b - VFCCU3 remains functional and is not impacted when CM7_3 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - VFCCU2 enters debug mode when CM7_3 enters debug mode. 1b - VFCCU2 remains functional and is not impacted when CM7_3 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when CM7_3 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when CM7_3 enters debug mode.

20.3.1.17 Read Write DEST 15 (GPR_RWD_15)

Offset

Register	Offset
GPR_RWD_15	103Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - STM3 remains functional and is not impacted when CM7_3 enters debug mode.
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - STM2 enters debug mode when CM7_3 enters debug mode. 1b - STM2 remains functional and is not impacted when CM7_3 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - STM1 enters debug mode when CM7_3 enters debug mode. 1b - STM1 remains functional and is not impacted when CM7_3 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - STM0 enters debug mode when CM7_3 enters debug mode. 1b - STM0 remains functional and is not impacted when CM7_3 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM7_3. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_3 enters debug mode when CM7_3 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when CM7_3 enters debug mode.

20.3.1.18 Read Write DEST 16 (GPR_RWD_16)

Offset

Register	Offset
GPR_RWD_16	1040h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - CPE_SWT0 remains functional and is not impacted when DSPV enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for DSPV. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when DSPV enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when DSPV enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for DSPV. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when DSPV enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when DSPV enters debug mode.

20.3.1.19 Read Write DEST 17 (GPR_RWD_17)

Offset

Register	Offset
GPR_RWD_17	1044h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for DSPV. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when DSPV enters debug mode. 1b - VFCCU1 remains functional and is not impacted when DSPV enters debug mode.

20.3.1.20 Read Write DEST 18 (GPR_RWD_18)

Offset

Register	Offset
GPR_RWD_18	1048h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															DEBU
W																G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-17 —	Reserved
16 DEBUG_DISABLE LE16	DEBUG_DISABLE Debug disable bit for DSPV. Set this bit 1 to disable the debug of module. 0b - SENT3 enters debug mode when DSPV enters debug mode. 1b - SENT3 remains functional and is not impacted when DSPV enters debug mode.

Table continues on the next page...

Table continued from the previous page...

Field	Function
DEBUG_DISAB LE0	0b - PIT_RTI_3 enters debug mode when DSPV enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when DSPV enters debug mode.

20.3.1.21 Read Write DEST 22 (GPR_RWD_22)

Offset

Register	Offset
GPR_RWD_22	1058h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserv	Reserv	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	ed	ed	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU	DEBU
W	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...	G_D...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 —	Reserved
30 —	Reserved
29 DEBUG_DISAB LE29	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - LCU1 enters debug mode when CM4 enters debug mode.

Table continues on the next page...

Table continued from the previous page...

Field	Function
5 DEBUG_DISABLE LE5	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - CPE_PIT0 enters debug mode when CM4 enters debug mode. 1b - CPE_PIT0 remains functional and is not impacted when CM4 enters debug mode.
4 DEBUG_DISABLE LE4	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - CPE_VFCCU enters debug mode when CM4 enters debug mode. 1b - CPE_VFCCU remains functional and is not impacted when CM4 enters debug mode.
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - CPE_SWT1 enters debug mode when CM4 enters debug mode. 1b - CPE_SWT1 remains functional and is not impacted when CM4 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - CPE_SWT0 enters debug mode when CM4 enters debug mode. 1b - CPE_SWT0 remains functional and is not impacted when CM4 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when CM4 enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when CM4 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when CM4 enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when CM4 enters debug mode.

20.3.1.22 Read Write DEST 23 (GPR_RWD_23)

Offset

Register	Offset
GPR_RWD_23	105Ch

Table continued from the previous page...

Field	Function
	1b - DSPI1 remains functional and is not impacted when CM4 enters debug mode.
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - DSPI0 enters debug mode when CM4 enters debug mode. 1b - DSPI0 remains functional and is not impacted when CM4 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - VFCCU3 enters debug mode when CM4 enters debug mode. 1b - VFCCU3 remains functional and is not impacted when CM4 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - VFCCU2 enters debug mode when CM4 enters debug mode. 1b - VFCCU2 remains functional and is not impacted when CM4 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when CM4 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when CM4 enters debug mode.

20.3.1.23 Read Write DEST 24 (GPR_RWD_24)

Offset

Register	Offset
GPR_RWD_24	1060h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - STM3 remains functional and is not impacted when CM4 enters debug mode.
3 DEBUG_DISAB LE3	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - STM2 enters debug mode when CM4 enters debug mode. 1b - STM2 remains functional and is not impacted when CM4 enters debug mode.
2 DEBUG_DISAB LE2	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - STM1 enters debug mode when CM4 enters debug mode. 1b - STM1 remains functional and is not impacted when CM4 enters debug mode.
1 DEBUG_DISAB LE1	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - STM0 enters debug mode when CM4 enters debug mode. 1b - STM0 remains functional and is not impacted when CM4 enters debug mode.
0 DEBUG_DISAB LE0	DEBUG_DISABLE Debug disable bit for CM4. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_3 enters debug mode when CM4 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when CM4 enters debug mode.

20.3.1.24 Read Write DEST 25 (GPR_RWD_25)

Offset

Register	Offset
GPR_RWD_25	1064h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - CPE_SWT1 enters debug mode when Cortex-R52_0 enters debug mode. 1b - CPE_SWT1 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - CPE_SWT0 enters debug mode when Cortex-R52_0 enters debug mode. 1b - CPE_SWT0 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when Cortex-R52_0 enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when Cortex-R52_0 enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when Cortex-R52_0 enters debug mode.

20.3.1.25 Read Write DEST 26 (GPR_RWD_26)

Offset

Register	Offset
GPR_RWD_26	1068h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - DSPI1 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
3 DEBUG_DISABLE LE3	Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - DSPI0 enters debug mode when Cortex-R52_0 enters debug mode. 1b - DSPI0 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
2 DEBUG_DISABLE LE2	Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - VFCCU3 enters debug mode when Cortex-R52_0 enters debug mode. 1b - VFCCU3 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
1 DEBUG_DISABLE LE1	Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - VFCCU2 enters debug mode when Cortex-R52_0 enters debug mode. 1b - VFCCU2 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
0 DEBUG_DISABLE LE0	Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when Cortex-R52_0 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when Cortex-R52_0 enters debug mode.

20.3.1.26 Read Write DEST 27 (GPR_RWD_27)

Offset

Register	Offset
GPR_RWD_27	106Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	0b - STM3 enters debug mode when Cortex-R52_0 enters debug mode. 1b - STM3 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
3 DEBUG_DISAB LE3	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - STM2 enters debug mode when Cortex-R52_0 enters debug mode. 1b - STM2 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
2 DEBUG_DISAB LE2	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - STM1 enters debug mode when Cortex-R52_0 enters debug mode. 1b - STM1 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
1 DEBUG_DISAB LE1	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - STM0 enters debug mode when Cortex-R52_0 enters debug mode. 1b - STM0 remains functional and is not impacted when Cortex-R52_0 enters debug mode.
0 DEBUG_DISAB LE0	DEBUG_DISABLE Debug disable bit for Cortex-R52_0. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_3 enters debug mode when Cortex-R52_0 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when Cortex-R52_0 enters debug mode.

20.3.1.27 Read Write DEST 28 (GPR_RWD_28)

Offset

Register	Offset
GPR_RWD_28	1070h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
3 DEBUG_DISABLE LE3	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - CPE_SWT1 enters debug mode when Cortex-R52_1 enters debug mode. 1b - CPE_SWT1 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
2 DEBUG_DISABLE LE2	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - CPE_SWT0 enters debug mode when Cortex-R52_1 enters debug mode. 1b - CPE_SWT0 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
1 DEBUG_DISABLE LE1	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - CPE_STM1 enters debug mode when Cortex-R52_1 enters debug mode. 1b - CPE_STM1 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
0 DEBUG_DISABLE LE0	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - CPE_STM0 enters debug mode when Cortex-R52_1 enters debug mode. 1b - CPE_STM0 remains functional and is not impacted when Cortex-R52_1 enters debug mode.

20.3.1.28 Read Write DEST 29 (GPR_RWD_29)

Offset

Register	Offset
GPR_RWD_29	1074h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - DSPI1 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
3 DEBUG_DISABLE LE3	Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - DSPI0 enters debug mode when Cortex-R52_1 enters debug mode. 1b - DSPI0 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
2 DEBUG_DISABLE LE2	Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - VFCCU3 enters debug mode when Cortex-R52_1 enters debug mode. 1b - VFCCU3 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
1 DEBUG_DISABLE LE1	Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - VFCCU2 enters debug mode when Cortex-R52_1 enters debug mode. 1b - VFCCU2 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
0 DEBUG_DISABLE LE0	Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - VFCCU1 enters debug mode when Cortex-R52_1 enters debug mode. 1b - VFCCU1 remains functional and is not impacted when Cortex-R52_1 enters debug mode.

20.3.1.29 Read Write DEST 30 (GPR_RWD_30)

Offset

Register	Offset
GPR_RWD_30	1078h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	0b - STM3 enters debug mode when Cortex-R52_1 enters debug mode. 1b - STM3 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
3 DEBUG_DISAB LE3	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - STM2 enters debug mode when Cortex-R52_1 enters debug mode. 1b - STM2 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
2 DEBUG_DISAB LE2	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - STM1 enters debug mode when Cortex-R52_1 enters debug mode. 1b - STM1 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
1 DEBUG_DISAB LE1	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - STM0 enters debug mode when Cortex-R52_1 enters debug mode. 1b - STM0 remains functional and is not impacted when Cortex-R52_1 enters debug mode.
0 DEBUG_DISAB LE0	DEBUG_DISABLE Debug disable bit for Cortex-R52_1. Set this bit 1 to disable the debug of module. 0b - PIT_RTI_3 enters debug mode when Cortex-R52_1 enters debug mode. 1b - PIT_RTI_3 remains functional and is not impacted when Cortex-R52_1 enters debug mode.

20.3.1.30 Read Write DEST 34 (GPR_RWD_34)

Offset

Register	Offset
GPR_RWD_34	1088h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - Fault routing enabled.
2 FAULT_CH_A_EN2	Enable routing of current injection fault. Enable routing of current injection fault for GPIO82 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
1 FAULT_CH_A_EN1	Enable routing of current injection fault. Enable routing of current injection fault for GPIO81 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
0 FAULT_CH_A_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO80 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.31 Read Write DEST 35 (GPR_RWD_35)

Offset

Register	Offset
GPR_RWD_35	108Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
0 FAULT_CH_A_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO126 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.32 Read Write DEST 36 (GPR_RWD_36)

Offset

Register	Offset
GPR_RWD_36	1090h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 FAULT_CH_A_EN31	Enable routing of current injection fault. Enable routing of current injection fault for GPIO215 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
30	Enable routing of current injection fault. Enable routing of current injection fault for GPIO115 on NCF[99] of VFCCU_1.

Table continues on the next page...

20.3.1.33 Read Write DEST 37 (GPR_RWD_37)

Offset

Register	Offset
GPR_RWD_37	1094h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 FAULT_CH_A_EN31	Enable routing of current injection fault. Enable routing of current injection fault for GPIO245 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
30 FAULT_CH_A_EN30	Enable routing of current injection fault. Enable routing of current injection fault for GPIO244 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
29 FAULT_CH_A_EN29	Enable routing of current injection fault. Enable routing of current injection fault for GPIO243 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
28	Enable routing of current injection fault.

Table continues on the next page...

Table continued from the previous page...

Field	Function
FAULT_CH_A_EN5	0b - Fault routing disabled. 1b - Fault routing enabled.
4 FAULT_CH_A_EN4	Enable routing of current injection fault. Enable routing of current injection fault for GPIO214 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
3 FAULT_CH_A_EN3	Enable routing of current injection fault. Enable routing of current injection fault for GPIO213 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
2 FAULT_CH_A_EN2	Enable routing of current injection fault. Enable routing of current injection fault for GPIO212 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
1 FAULT_CH_A_EN1	Enable routing of current injection fault. Enable routing of current injection fault for GPIO209 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
0 FAULT_CH_A_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO208 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.34 Read Write DEST 38 (GPR_RWD_38)

Offset

Register	Offset
GPR_RWD_38	1098h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - Fault routing enabled.
2 FAULT_CH_A_EN2	Enable routing of current injection fault. Enable routing of current injection fault for GPIO249 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
1 FAULT_CH_A_EN1	Enable routing of current injection fault. Enable routing of current injection fault for GPIO247 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
0 FAULT_CH_A_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO246 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.35 Read Write DEST 39 (GPR_RWD_39)

Offset

Register	Offset
GPR_RWD_39	109Ch

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
0 FAULT_CH_B_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO232 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.36 Read Write DEST 40 (GPR_RWD_40)

Offset

Register	Offset
GPR_RWD_40	10A0h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 FAULT_CH_A_EN31	Enable routing of current injection fault. Enable routing of current injection fault for GPIO328 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
30	Enable routing of current injection fault. Enable routing of current injection fault for GPIO327 on NCF[99] of VFCCU_1.

Table continues on the next page...

20.3.1.37 Read Write DEST 41 (GPR_RWD_41)

Offset

Register	Offset
GPR_RWD_41	10A4h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 FAULT_CH_B_EN31	Enable routing of current injection fault. Enable routing of current injection fault for GPIO250 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
30 FAULT_CH_B_EN30	Enable routing of current injection fault. Enable routing of current injection fault for GPIO248 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
29 FAULT_CH_B_EN29	Enable routing of current injection fault. Enable routing of current injection fault for GPIO315 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
28	Enable routing of current injection fault.

Table continues on the next page...

Table continued from the previous page...

Field	Function
FAULT_CH_A_EN5	0b - Fault routing disabled. 1b - Fault routing enabled.
4 FAULT_CH_A_EN4	Enable routing of current injection fault. Enable routing of current injection fault for GPIO333 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
3 FAULT_CH_A_EN3	Enable routing of current injection fault. Enable routing of current injection fault for GPIO332 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
2 FAULT_CH_A_EN2	Enable routing of current injection fault. Enable routing of current injection fault for GPIO331 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
1 FAULT_CH_A_EN1	Enable routing of current injection fault. Enable routing of current injection fault for GPIO330 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
0 FAULT_CH_A_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO329 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.38 Read Write DEST 42 (GPR_RWD_42)

Offset

Register	Offset
GPR_RWD_42	10A8h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
	1b - Fault routing enabled.
2 FAULT_CH_B_EN2	Enable routing of current injection fault. Enable routing of current injection fault for GPIO264 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
1 FAULT_CH_B_EN1	Enable routing of current injection fault. Enable routing of current injection fault for GPIO255 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
0 FAULT_CH_B_EN0	Enable routing of current injection fault. Enable routing of current injection fault for GPIO252 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.

20.3.1.39 Read Write DEST 43 (GPR_RWD_43)

Offset

Register	Offset
GPR_RWD_43	10ACh

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...
W		FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...	FAULT_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

20.3.1.40 Read Write DEST 40 (GPR_RWD_45)

Offset

Register	Offset
GPR_RWD_45	10B4h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W									_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-25 —	Reserved
24 FAULT_CH_A_EN24	Enable routing of current injection fault. Enable routing of current injection fault for GPIO371 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
23 FAULT_CH_A_EN23	Enable routing of current injection fault. Enable routing of current injection fault for GPIO370 on NCF[99] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
22 FAULT_CH_A_EN22	Enable routing of current injection fault. Enable routing of current injection fault for GPIO369 on NCF[99] of VFCCU_1. 0b - Fault routing disabled.

Table continues on the next page...

20.3.1.41 Read Write DEST 42 (GPR_RWD_46)

Offset

Register	Offset
GPR_RWD_46	10B8h

Function

This register functions on DEST reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W									_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT	FAULT
W	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...	_C...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-25 —	Reserved
24 FAULT_CH_B_EN24	Enable routing of current injection fault. Enable routing of current injection fault for GPIO371 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
23 FAULT_CH_B_EN23	Enable routing of current injection fault. Enable routing of current injection fault for GPIO370 on NCF[100] of VFCCU_1. 0b - Fault routing disabled. 1b - Fault routing enabled.
22 FAULT_CH_B_EN22	Enable routing of current injection fault. Enable routing of current injection fault for GPIO369 on NCF[100] of VFCCU_1. 0b - Fault routing disabled.

Table continues on the next page...

20.3.1.42 Read Only DEST 0 (GPR_ROD_0)

Offset

Register	Offset
GPR_ROD_0	1800h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 CURRENT_INJ_ACTIVE31	Current injection active status bit. Current is being actively injected on GPIO116. 0b - Not present. 1b - Present.
30 CURRENT_INJ_ACTIVE30	Current injection active status bit. Current is being actively injected on GPIO112. 0b - Not present. 1b - Present.
29 CURRENT_INJ_ACTIVE29	Current injection active status bit. Current is being actively injected on GPIO111. 0b - Not present. 1b - Present.

Table continues on the next page...

Table continued from the previous page...

Field	Function
CURRENT_INJ_ACTIVE5	Current is being actively injected on GPIO85. 0b - Not present. 1b - Present.
4 CURRENT_INJ_ACTIVE4	Current injection active status bit. Current is being actively injected on GPIO84. 0b - Not present. 1b - Present.
3 CURRENT_INJ_ACTIVE3	Current injection active status bit. Current is being actively injected on GPIO83. 0b - Not present. 1b - Present.
2 CURRENT_INJ_ACTIVE2	Current injection active status bit. Current is being actively injected on GPIO82. 0b - Not present. 1b - Present.
1 CURRENT_INJ_ACTIVE1	Current injection active status bit. Current is being actively injected on GPIO81. 0b - Not present. 1b - Present.
0 CURRENT_INJ_ACTIVE0	Current injection active status bit. Current is being actively injected on GPIO80. 0b - Not present. 1b - Present.

20.3.1.43 Read Only DEST 1 (GPR_ROD_1)

Offset

Register	Offset
GPR_ROD_1	1804h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
CURRENT_INJ_ACTIVE3	0b - Not present. 1b - Present.
2 CURRENT_INJ_ACTIVE2	Current injection active status bit. Current is being actively injected on GPIO128. 0b - Not present. 1b - Present.
1 CURRENT_INJ_ACTIVE1	Current injection active status bit. Current is being actively injected on GPIO127. 0b - Not present. 1b - Present.
0 CURRENT_INJ_ACTIVE0	Current injection active status bit. Current is being actively injected on GPIO126. 0b - Not present. 1b - Present.

20.3.1.44 Read Only DEST 2 (GPR_ROD_2)

Offset

Register	Offset
GPR_ROD_2	1808h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
CURRENT_INJ_ACTIVE3	0b - Not present. 1b - Present.
2 CURRENT_INJ_ACTIVE2	Current injection active status bit. Current is being actively injected on GPIO235. 0b - Not present. 1b - Present.
1 CURRENT_INJ_ACTIVE1	Current injection active status bit. Current is being actively injected on GPIO234. 0b - Not present. 1b - Present.
0 CURRENT_INJ_ACTIVE0	Current injection active status bit. Current is being actively injected on GPIO233. 0b - Not present. 1b - Present.

20.3.1.45 Read Only DEST 3 (GPR_ROD_3)

Offset

Register	Offset
GPR_ROD_3	180Ch

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
CURRENT_INJ_ACTIVE3	0b - Not present. 1b - Present.
2 CURRENT_INJ_ACTIVE2	Current injection active status bit. Current is being actively injected on GPIO311. 0b - Not present. 1b - Present.
1 CURRENT_INJ_ACTIVE1	Current injection active status bit. Current is being actively injected on GPIO310. 0b - Not present. 1b - Present.
0 CURRENT_INJ_ACTIVE0	Current injection active status bit. Current is being actively injected on GPIO309. 0b - Not present. 1b - Present.

20.3.1.46 Read Only DEST 4 (GPR_ROD_4)

Offset

Register	Offset
GPR_ROD_4	1810h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
CURRENT_INJ_ACTIVE3	0b - Not present. 1b - Present.
2 CURRENT_INJ_ACTIVE2	Current injection active status bit. Current is being actively injected on GPIO343. 0b - Not present. 1b - Present.
1 CURRENT_INJ_ACTIVE1	Current injection active status bit. Current is being actively injected on GPIO342. 0b - Not present. 1b - Present.
0 CURRENT_INJ_ACTIVE0	Current injection active status bit. Current is being actively injected on GPIO341. 0b - Not present. 1b - Present.

20.3.1.47 Read Only DEST 5 (GPR_ROD_5)

Offset

Register	Offset
GPR_ROD_5	1814h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
CURRENT_INJ_COMPLETE3	0b - Current injection has not occurred. 1b - Current injection has occurred.
2 CURRENT_INJ_COMPLETE2	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO93. 0b - Current injection has not occurred. 1b - Current injection has occurred.
1 CURRENT_INJ_COMPLETE1	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO91. 0b - Current injection has not occurred. 1b - Current injection has occurred.
0 CURRENT_INJ_COMPLETE0	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO90. 0b - Current injection has not occurred. 1b - Current injection has occurred.

20.3.1.48 Read Only DEST 6 (GPR_ROD_6)

Offset

Register	Offset
GPR_ROD_6	1818h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
2 CURRENT_INJ_COMPLETE2	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO137. 0b - Current injection has not occurred. 1b - Current injection has occurred.
1 CURRENT_INJ_COMPLETE1	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO136. 0b - Current injection has not occurred. 1b - Current injection has occurred.
0 CURRENT_INJ_COMPLETE0	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO135. 0b - Current injection has not occurred. 1b - Current injection has occurred.

20.3.1.49 Read Only DEST 7 (GPR_ROD_7)

Offset

Register	Offset
GPR_ROD_7	181Ch

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CURR ENT...	CURR ENT...	Reserv ed	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W	W1C	W1C		W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
CURRENT_INJ_COMPLETE0	0b - Current injection has not occurred. 1b - Current injection has occurred.

20.3.1.50 Read Only DEST 8 (GPR_ROD_8)

Offset

Register	Offset
GPR_ROD_8	1820h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 CURRENT_INJ_COMPLETE31	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO349. 0b - Current injection has not occurred. 1b - Current injection has occurred.
30 CURRENT_INJ_COMPLETE30	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO348.

Table continues on the next page...

20.3.1.51 Read Only DEST 9 (GPR_ROD_9)

Offset

Register	Offset
GPR_ROD_9	1824h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved		CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...	CURR ENT...
W			W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-14 —	Reserved
13 CURRENT_INJ_COMPLETE13	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO363. 0b - Current injection has not occurred. 1b - Current injection has occurred.
12 CURRENT_INJ_COMPLETE12	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO362. 0b - Current injection has not occurred. 1b - Current injection has occurred.
11 CURRENT_INJ_COMPLETE11	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO361.

Table continues on the next page...

Table continued from the previous page...

Field	Function
CURRENT_INJ_COMPLETE3	This flag latches the current injection information for GPIO353. 0b - Current injection has not occurred. 1b - Current injection has occurred.
2 CURRENT_INJ_COMPLETE2	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO352. 0b - Current injection has not occurred. 1b - Current injection has occurred.
1 CURRENT_INJ_COMPLETE1	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO351. 0b - Current injection has not occurred. 1b - Current injection has occurred.
0 CURRENT_INJ_COMPLETE0	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO350. 0b - Current injection has not occurred. 1b - Current injection has occurred.

20.3.1.52 Read Only DEST 4 (GPR_ROD_11)

Offset

Register	Offset
GPR_ROD_11	1848h

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
4 CURRENT_INJ_ACTIVE4	Current injection active status bit. Current is being actively injected on GPIO115. 0b - Not present. 1b - Present.
3 CURRENT_INJ_ACTIVE3	Current injection active status bit. Current is being actively injected on GPIO114. 0b - Not present. 1b - Present.
2 CURRENT_INJ_ACTIVE2	Current injection active status bit. Current is being actively injected on GPIO113. 0b - Not present. 1b - Present.
1 CURRENT_INJ_ACTIVE1	Current injection active status bit. Current is being actively injected on GPIO92. 0b - Not present. 1b - Present.
0 CURRENT_INJ_ACTIVE0	Current injection active status bit. Current is being actively injected on GPIO88. 0b - Not present. 1b - Present.

20.3.1.53 Read Only DEST 5 (GPR_ROD_12)

Offset

Register	Offset
GPR_ROD_12	184Ch

Function

This register functions on DEST reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
4 CURRENT_INJ_COMPLETE4	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO115. 0b - Current injection has not occurred. 1b - Current injection has occurred.
3 CURRENT_INJ_COMPLETE3	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO114. 0b - Current injection has not occurred. 1b - Current injection has occurred.
2 CURRENT_INJ_COMPLETE2	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO113. 0b - Current injection has not occurred. 1b - Current injection has occurred.
1 CURRENT_INJ_COMPLETE1	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO92. 0b - Current injection has not occurred. 1b - Current injection has occurred.
0 CURRENT_INJ_COMPLETE0	Flag indicating that current injection occurred on a GPIO. This flag latches the current injection information for GPIO88. 0b - Current injection has not occurred. 1b - Current injection has occurred.

20.3.1.54 Read Write FUNC 0 (GPR_RWF_0)

Offset

Register	Offset
GPR_RWF_0	2000h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
CORE1_FD_RE Q	This bit isolates the CM7_1 interface from NIC_SYS. This isolation needs to be asserted before functional reset is asserted in the device. 0b - No isolation. 1b - Isolation enabled.
1 CORE0_FD_RE Q	CM7_0 isolation request signal. This bit isolates the CM7_0 interface from NIC_SYS. This isolation needs to be asserted before functional reset is asserted in the device. 0b - No isolation. 1b - Isolation enabled.
0 AXI_ACC_POR T_FD_REQ	Fence and drain request signal When this bit is asserted, this signals the controller gasket to begin fence-and-drain operations. 0b - The AXI signals flow through the controller gasket unchanged. 1b - The controller gasket overrides the controller agent and directly drives the AXI interface to block new transactions issued by the controller from being propagated downstream and issue all outstanding AXI write commands and write data on behalf of the controller.

20.3.1.55 Read Write FUNC 1 (GPR_RWF_1)

Offset

Register	Offset
GPR_RWF_1	2004h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved		TIME_	TIME_	TIME_	TIME_	TIME_	TIME_	TIME_	TIME_	TIME_	Reserved				
W			BA...	BA...	BA...	BA...	BA...	BA...	BA...	BA...	BA...					
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CURR_	NPU_	Reserv	Reserv	SSI3_I	Reserv	Reserv	SSI2_I	Reserv	Reserv	SSI1_I	Reserv	SSI0_I	Reserv	SSI0_	SSI0_
W	_IN...	_DIS...	ed	ed	S...	ed	ed	S...	ed	ed	S...	ed	S...	ed	PA...	BL...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

20.3.1.56 Read Write FUNC 2 (GPR_RWF_2)

Offset

Register	Offset
GPR_RWF_2	2008h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved														RTU_S	CAN_T
W															AF...	S...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_	CAN_
W	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...	TS...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-18 —	Reserved
17 RTU_SAFE_CLOCK_GATING_EN	RTU Safe clock gating enable bit. Controlling safe clock gating when only RTU is going through func reset. 0b - Disabled the clock gating. 1b - Enabled the clock gating.
16 CAN_TS_SEL16	FlexCAN_16 timestamp select bit. Select between TMR_1588 and STM0 for FlexCAN_16 timestamping 0b - TMR_1588 selected for FlexCAN_16 timestamping. 1b - STM0 selected for FlexCAN_16 timestamping.
15 CAN_TS_SEL15	FlexCAN_15 timestamp select bit. Select between TMR_1588 and STM0 for FlexCAN_15 timestamping 0b - TMR_1588 selected for FlexCAN_15 timestamping.

Table continues on the next page...

20.3.1.57 Read Write FUNC 3 (GPR_RWF_3)

Offset

Register	Offset
GPR_RWF_3	200Ch

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	SSI3_	SSI3_	SSI3_	INJ_SM_FAULT_SM_SS			SSI3_	SSI3_	SSI2_	SSI2_	SSI2_	INJ_SM_FAULT_SM_SS			SSI2_	SSI2_
W	ER...	BI...	ER...	I3			FA...	CL...	ER...	BI...	ER...	I2			FA...	CL...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	SSI1_	SSI1_	SSI1_	INJ_SM_FAULT_SM_SS			SSI1_	SSI1_	SSI0_	SSI0_	SSI0_	INJ_SM_FAULT_SM_SS			SSI0_	SSI0_
W	ER...	BI...	ER...	I1			FA...	CL...	ER...	BI...	ER...	I0			FA...	CL...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 SSI3_ERR_INJ	APB write to error injection registers. APB write to error injection registers of SSI3. 0b - APB write to error injection registers will be ignored. 1b - APB write to error injection registers will inject a parity error or mission fault on corresponding signal.
30 SSI3_BIST_TEST	Assert to use APB test registers for BIST testing. Assert to use APB test registers for BIST testing for SSI3.
29 SSI3_ERR_CLEAR	ASSERT_CLEAR Assert to clear ssi_parity_error,ssi_parity_etype,mission_fault, sm_fault_or, pwr_ctrl_pe_sync,slice_lost, and ctrl_logic_failure status signal of SSI3.
28-26 INJ_SM_FAULT_SM_SSI3	Set correct value for this signal before pulsing inj_sm_fault signal. This signal represents following state machines of SSI3: 000b - Channel Arbiter.

Table continues on the next page...

Table continued from the previous page...

Field	Function
SSI0_ERR_INJ	0b - APB write to error injection registers will be ignored. 1b - APB write to error injection registers will inject a parity error or mission fault on corresponding signal.
6 SSI0_BIST_TEST	Assert to use APB test registers for BIST testing. Assert to use APB test registers for BIST testing for SSI0.
5 SSI0_ERR_CLEAR	ASSERT_CLEAR Assert to clear ssi_parity_error,ssi_parity_etype,mission_fault, sm_fault_or, pwr_ctrl_pe_sync,slice_lost, and ctrl_logic_failure status signal of SSI0.
4-2 INJ_SM_FAULT_SM_SSI0	Set correct value for this signal before pulsing inj_sm_fault signal. This signal represents following state machines of SSI0: 000b - Channel Arbiter. 001b - Power Down. 010b - Initiator : Primary Read Address BHM, Responder : Reserved. 011b - Initiator : Primary Write Address BHM, Responder : Reserved. 100b - Auxiliary Read Address BHM; do only applicable to AUX NONE variants. 101b - Auxiliary Write Address BHM; do only applicable to AUX NONE variants.
1 SSI0_FAULT_INJ	SELECT_FAULT_INJECTION Selects where to insert fault between primary control logic and secondary control logic of SSI0. 0b - Inject fault in primary control logic. 1b - Inject fault in secondary control logic.
0 SSI0_CLR_UNINT	Assert this signal to clear unint_mode_trans_det status output of SSI0. Assert this signal to clear unint_mode_trans_det status output of SSI0.

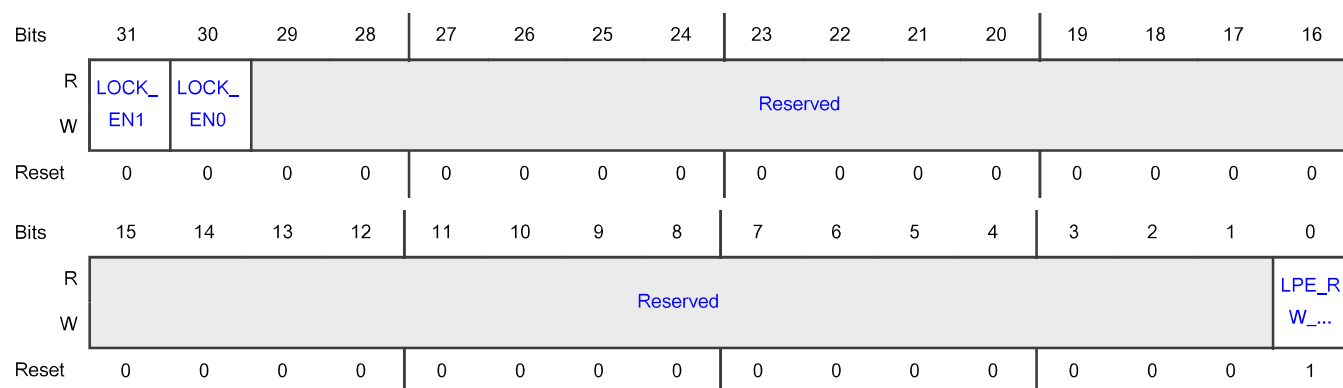
20.3.1.58 Read Write FUNC 4 (GPR_RWF_4)

Offset

Register	Offset
GPR_RWF_4	2010h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram**Fields**

Field	Function
31 LOCK_EN1	Lock enable. If LOCK_EN1 and LOCK_EN0 both are 0 then register can be written. If LOCK_EN1 is 0 and LOCK_EN0 is 1 then register can be written. If LOCK_EN1 is 1 and LOCK_EN0 is 0 then register is temporarily protected. If LOCK_EN1 and LOCK_EN0 both are 1 then this register is locked (read-only) until the next reset.
30 LOCK_EN0	Lock enable. For detailed description of LOCK_EN0 and LOCK_EN1 bit please see LOCK_EN1 section.
29-1 —	Reserved
0 LPE_RW_BLOCK_EN	Read write blocking bit. This bit blocks the read and write from LPE domain to RUN domain. Any read and write transaction from LPE domain to RUN domain will be blocked if this bit is set to 1. 0b - Blocking logic disabled 1b - Blocking logic enabled

20.3.1.59 Read Write FUNC 5 (GPR_RWF_5)**Offset**

Register	Offset
GPR_RWF_5	2014h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

20.3.1.60 Read Write FUNC 6 (GPR_RWF_6)

Offset

Register	Offset
GPR_RWF_6	2018h

Function

This register is used by application core to communicate with HSE2. This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
R	APP_H	APP_H	APP_H	APP_H	APP_H					APP_H	APP_H	APP_H	APP_H	APP_H	APP_H		
W	SE...	SE...	SE...	SE...	SE...	Reserved				SE...	SE...	SE...	SE...	SE...	SE...	Reserved	
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
R	Reserved								APP_H	Reserved							
W									SE...								
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Fields

Field	Function
31 APP_HSE2_CO MM_31	APPLICATION_HSE2_COMM_31. The application core writes to this bit to communicate with HSE2. Refer to HSE2 Reference manual for details. 0b - No communication initialized by application core with HSE2. 1b - Application core initialized communication with HSE2.
30 APP_HSE2_CO MM_30	APPLICATION_HSE2_COMM_30. The application core writes to this bit to communicate with HSE2. Refer to HSE2 Reference manual for details. 0b - No communication initialized by application core with HSE2. 1b - Application core initialized communication with HSE2.
29 APP_HSE2_CO MM_29	APPLICATION_HSE2_COMM_29. The application core writes to this bit to communicate with HSE2. Refer to HSE2 Reference manual for details. 0b - No communication initialized by application core with HSE2.

Table continues on the next page...

Table continued from the previous page...

Field	Function
19 APP_HSE2_CO MM_19	APPLICATION_HSE2_COMM_19. The application core writes to this bit to communicate with HSE2. Refer to HSE2 Reference manual for details. 0b - No communication initialized by application core with HSE2. 1b - Application core initialized communication with HSE2.
18 APP_HSE2_CO MM_18	APPLICATION_HSE2_COMM_18. The application core writes to this bit to communicate with HSE2. Refer to HSE2 Reference manual for details. 0b - No communication initialized by application core with HSE2. 1b - Application core initialized communication with HSE2.
17-8 —	Reserved
7 APP_HSE2_CO MM_7	APPLICATION_HSE2_COMM_7. The application core writes to this bit to communicate with HSE2. Refer to HSE2 Reference manual for details. 0b - No communication initialized by application core with HSE2. 1b - Application core initialized communication with HSE2.
6-0 —	Reserved

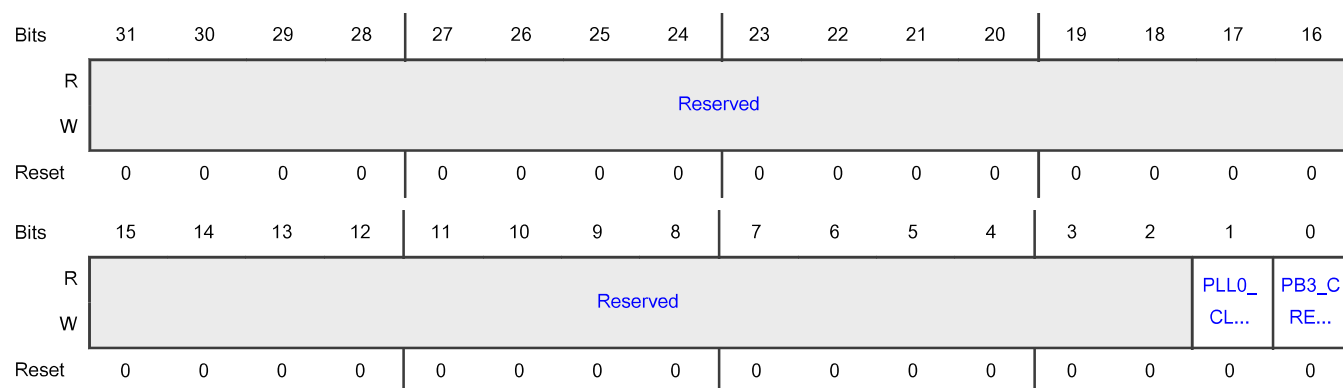
20.3.1.61 Read Write FUNC 7 (GPR_RWF_7)

Offset

Register	Offset
GPR_RWF_7	201Ch

Function

This register functions on Func reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Diagram**Fields**

Field	Function
31-2 —	Reserved
1 PLL0_CLK_DIVIDER_DISABLE	To disable the PLL0 clock divider. It is used to program idle bit of PLL0_CLK divider to disable the divider before gating PLL0_CLK. It helps to gate output of divider at it's low phase. 0b - Divider is not disabled. 1b - Divider is disabled.
0 PB3_CREF_EN	CREF enable. PBRIDGE3 CREF Enable.

20.3.1.62 Read Write FUNC 3 (GPR_RWF_8)**Offset**

Register	Offset
GPR_RWF_8	2020h

Function

This register functions on FUNC reset and its access type is RW. The details of function of each bit of this register is specified in the table below.

Table continued from the previous page...

Field	Function
0	WFI Bypass Acknowledge bit for R52_0 Core.
WFI_BYPASS_ACK_R52_0	0b - WFI bypass for R52_0_CORE Disabled for LPRUN/Standby entry. 1b - WFI bypass for R52_0_CORE Enabled for LPRUN/Standby entry.

20.3.1.63 Read Only FUNC 0 (GPR_ROF_0)

Offset

Register	Offset
GPR_ROF_0	2800h

Function

This register functions on FUNC reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved	AES_RESULT_DID_ERR_DID				AES_RES...	AES_RES...	AES_FEED_DID_ERR_DID				AES_FEE...	AES_FEE...	AES_SAFETY_ERR_DID		
W		W1C				W1C	W1C	W1C				W1C	W1C	W1C		
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	AES_SAF...	AES_SAF...	AES_SAF...	Reserved				ETR_FD...	CORE3_F...	CORE2_F...	CORE1_F...	CORE0_F...	AXI_ACC...	SSI3_ID...	SSI2_ID...	SSI1_ID...
W	W1C	W1C	W1C													
Reset	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

Fields

Field	Function
31	Reserved
—	
30-27	This register bit indicates the DID[3:0] value when the AES result DID error was reported.
AES_RESULT_DID_ERR_DID	This register bit indicates the DID[3:0] value when the AES result DID error was reported.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Indicates that all primary AXI transactions through the SSI module have not finished. 1b - Indicates that all primary AXI transactions through the SSI module have finished.
2 SSI2_IDLE	SSI2 IDLE SSI2 IDLE 0b - Indicates that all primary AXI transactions through the SSI module have not finished. 1b - Indicates that all primary AXI transactions through the SSI module have finished.
1 SSI1_IDLE	SSI1 IDLE SSI1 IDLE 0b - Indicates that all primary AXI transactions through the SSI module have not finished. 1b - Indicates that all primary AXI transactions through the SSI module have finished.
0 SSI0_IDLE	SSI0 IDLE SSI0 IDLE 0b - Indicates that all primary AXI transactions through the SSI module have not finished. 1b - Indicates that all primary AXI transactions through the SSI module have finished.

20.3.1.64 Read Only FUNC 1 (GPR_ROF_1)

Offset

Register	Offset
GPR_ROF_1	2804h

Function

This register functions on FUNC reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved								RTU_S	RTU_S	RTU_S	RTU_S	INT_	INIT_O	CM7_3	CM7_2
W									SI...	SI...	SI...	SI...	PEND	C...	_G...	_G...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table continued from the previous page...

Field	Function
CM7_1_GSKT_HLT	This bit indicates when the AXI bus is idle, meaning there are no outstanding read nor write transactions. When a fence-and-drain request is issued, it indicates when all outstanding AXI transactions have been resolved, and the controller is properly quiesced.
0	controller idle indicator.
CM7_0_GSKT_HLT	This bit indicates when the AXI bus is idle, meaning there are no outstanding read nor write transactions. When a fence-and-drain request is issued, it indicates when all outstanding AXI transactions have been resolved, and the controller is properly quiesced.

20.3.1.65 Read Only FUNC 2 (GPR_ROF_2)

Offset

Register	Offset
GPR_ROF_2	2808h

Function

This register functions on FUNC reset and its access type is RO. The details of function of each bit of this register is specified in the table below.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved															PLL0_ CL...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Fields

Field	Function
31-1 —	Reserved
0	Acknowledgement of disablement of PLL0 clock divider.

Table continues on the next page...